

Diurnal Sorption-Condensation Engine (DSCE)

A Simple, Zero-Cost System to Pull Water from the Air

Original Creator: Subramanyam Yuvaraj • Open-Source CC-BY 4.0



1. What is this project?

This guide explains how to build a simple, 100% free system to harvest clean water directly from the air using everyday waste and scrap items. It is designed to help people get drinking water in emergency survival situations or remote places without spending any money.

Creator Note: I, Subramanyam Yuvaraj, created this design. It is shared freely with the world under an open-source license so anyone can build, test, and improve it.

2. Parts List (What you need)

You do not need to buy anything. Gather these discarded items instead:

- **Box:** An old Styrofoam shipping box or wooden crate to hold heat and structure.
- **Cover Window:** A scrap piece of window glass, picture frame glass, or clear plastic sheet.
- **Water Absorber:** Calcium chloride crystals from a store-bought dampness absorber bucket, clean wood ash, or silica gel packets.
- **Mesh Tray:** Pieces of a plastic fruit basket or old window screen mesh.
- **Dark Base Paint:** Crushed charcoal dust mixed with water to paint the inside bottom black so it absorbs sun heat quickly.
- **Vapor Pipe:** An empty ballpoint pen casing or small plastic tube.
- **Collection Bottle:** A clean waste plastic bottle kept in the shade or slightly buried.
- **Seals:** Sticky mud mixed with plant fibers or melted candle wax to keep the box airtight.

3. How it Works (Simple Steps)

The system works in a simple daily cycle:

- **Nighttime (Absorbing Moisture):** At night, open or vent the box to the cool air. The salt or absorber inside will pull moisture right out of the air ($\text{CaCl}_2 + n\text{H}_2\text{O} \rightarrow \text{CaCl}_2 \cdot n\text{H}_2\text{O}$).
- **Daytime (Making Steam):** In the morning, seal the box tightly with wax or clay and place it under direct sunlight. The inside gets very hot ($T > 55^\circ\text{C}$), forcing the trapped moisture out of the salt.
- **Collection:** The hot steam travels down the plastic tube, passes through a charcoal filter to clean it, and drips into your collection bottle as pure water.

4. Open-Source Rights

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Document ID: DSCE-2026-SR01

Safety & Water Purity Warning

Disclaimer: This project is an experimental concept meant for education and emergency survival only.

Do Not Drink Without Testing: Because this system uses scrap materials and chemicals like calcium chloride, **the water collected must NOT be drunk** until it has been tested in a professional laboratory for heavy metals, poisons, or germs. The creator takes no responsibility for any health issues or injuries from building or using this device.